

[DC] Situated augmented reality: beyond the egocentric viewpoint

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Figure 1: Situated visualization challenges: (a) Egocentric viewpoint limitation of the user [1], (b) Data overload [2], (c) Visual interference [3], (d) Visual coherence and registration errors [4], (e) Temporal coherence issues due to the dynamics of AR [2].

ABSTRACT

Augmented reality (AR) is well suited for situated visualization (SV), a method to represent data in context, with potential in many situations. Furthermore, AR-based SV poses challenges and research prospects. A vital one is the egocentric viewpoint limitation of the users, which reduces their ability to explore all the available information. To create new approaches that will overcome this limitation, this work began by understanding the relevance of viewpoints in typical AR-based SV scenarios, characterizing the approaches already proposed in order to tackle it. The new proposed approaches should be evaluated in different settings to improve and validate them, as well as propose guidelines for some relevant scenarios.

Index Terms: Augmented reality—situated visualization—egocentric viewpoint limitation

1 INTRODUCTION

When digital contents are merged with real-world environments, the information captured by the human senses is augmented, enhancing human perception of reality. This is referred to as augmented reality (AR) [5].

To take advantage of the user's context, supporting users while performing visualization tasks, AR applications may use situated visualization (SV), which comprises all the visualizations that change their appearance based on context, by considering the ones that are relevant to the physical context in which they are displayed [4]. According to [6], SV must be thought physically, perceptually, and temporally. A visualization is: "physically situated in space if its physical presentation is physically close to the data's physical referent", "perceptually situated in space if its percept (physical or virtual presentation) appears to be close to the percept of the data's physical referent", and "temporally situated if the data's temporal referent is close to the moment in time the physical presentation is observed". A physical referent is "a physical object or space to which the data refers". A physical presentation is "the physical apparatus that makes the visualization observable". The definitions' term "close" is left vague on purpose since situatedness is lying on a continuum with different levels. For example, a visualization projected on a physical referent is spatially more situated than a visualization seen on a mobile device near the physical referent. An important group for the closest SV possible is the embedded visualization, which "is the use of visual and physical representations of data that are deeply integrated with the physical spaces, objects, and entities to which the data refers" [7].

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Compared with other visualizations, SV offers high adaptability, usefulness, and intuitiveness by contextualizing the relevant information in the current physical space, leading to more informed decisions. Still, these benefits imply challenges that influence their applicability and utility, some due to the dynamic and distracting nature of the real-world. One of the main challenges is the egocentric viewpoint limitation of the AR user. The challenge is to see/collect data outside of the current viewpoint, avoiding or mitigating alterations of the user's position. Figure 1(a) illustrates the necessity of changing the user's field of view (FOV) to access more information outside the screen. To deal with this challenge, due to the necessity of handling additional information to create the new viewpoints, it is necessary to face other SV challenges, such as data overload (to provide the needed information, avoiding confusion and lack of clarity) [2], visual interference (to distinguish crucial information from the irrelevant one, avoiding the occlusion of vital data by the virtual content) [3], visual coherence (to deliver data that makes sense) [4], registration error (to overlay the digital content at the exact position) [4], dynamics of AR (to keep track of the changes done in the scenario or the user's viewpoint when digital content is merged, avoiding confusing outcomes) [2], and temporal coherence (to deliver data that makes sense at the exact time) [2]. All these challenges are typified in Figure 1. Tackling the egocentric viewpoint limitation could enhance user experience and safety, since the user can see more and become more aware of the surroundings.

2 EXTENDING THE AR EGOCENTRIC VIEWPOINT

Several techniques may be used to extend the AR user's egocentric viewpoint: multi-perspective renderings, transitional interfaces, off-screen visualizations, zooming and focus + context, and overview + detail [8].

Multi-perspective renderings add extra views of the real-world into the current AR user's egocentric viewpoint. This technique truly extends the egocentric viewpoint because users from the same location can visualize more relevant data around them. Some approaches of this technique are panoramic renderings, AR mirrors, and deformation procedures [8].

Transitional interface technique offers the users a seamless way to switch from the AR egocentric viewpoint to other relevant virtual reality (VR) viewpoints of the same elements without altering their position. These techniques can be divided into smart transitions (when supported by intelligent algorithms) and object-centric exploration (when full VR models of the real-world objects are used) [2].

The concept behind the off-screen visualizations is that the AR display is just a window into a larger space, making the user aware, by using 2D or 3D directional or neighbouring indications, that the desired information might not be visible at a given moment because it is out of the AR camera's FOV [8].

Overview + detail and zooming and focus + context techniques are used to level the viewpoint restriction of users when the workspace

is larger than the screen size [9]. In overview + detail, the current view of the user is seen as the detail, while other visualizations offer an overview of the entire area of interest. The zooming and focus + context, a combination of different techniques, establishes the detail view as a focus and then delivers the overview as a context that includes the focus area, using geometric and semantic zoom (freezes the real-time scene visualization, and applies different degrees of zoom according to the user's interest) [8].

According to [8], the typical device type used in the mitigation of this limitation are the mobile-based devices, although the video-see-through-based and other devices may also be employed.

3 PROPOSED WORK

The literature review in [8] suggests that the main weakness of the multi-perspective renderings techniques is the difficulty the users have in changing the viewpoint of the added images because they are usually static (specifically calculated for the situation). So, the idea is to start the work by coherently presenting and augmenting, in the user's display, real-time feeds of different static cameras, related to and relevant for the user's context. This process must detect if the feeds are of the same physical referents, separating/discarding the ones which are not (according with the task to be done) and creating panoramic views with the ones which are (providing the user with a dynamic larger FOV). Then, the same approach must be tried with mobile cameras and with scattered camera networks. It will be interesting to work with different types of information (for instance, digital, infrared, or time-of-flight images), particularly when creating the panoramic views. With this the user could be situated in a broader area, having the feeling of "being" in different places, at the same time, while seeing distinct views of the physical referents. The goal is to change the viewpoint using the knowledge of where the user wants or must look at (or, in other words, which view must be presented), allowing an effective improvement in the use of AR-based SV systems. This will imply the use of artificial intelligence (AI), tracking and recognition procedures. This work must address critical issues, such as the size of the display as well as the processing capacity of the used device, and deal with additional challenges on how to retrieve, combine, synchronize, and present all the gathered data in order to optimize the tasks to be done.

Different to what others have done [2], the next step will be the study on the possibility of using transitional interfaces techniques and multi-perspective renderings jointly. This work must begin by merging virtual camera feeds with real ones, in known and controlled environments (due to the need of having a virtual model of the real-world scenario where the user is). It must be complemented with a study of where to insert visual cues to reduce the task conclusion time and the perceived task load (in a similar way as off-screen visualization techniques). It is an opportunity to use the semantic zoom technique due to the spread of AI, because semantic information might be used to strategically present relevant viewpoints according to the user's current task and context, considering, for example, smart transitional interfaces that detect and adapt to undesired occlusions, from certain views. The intelligent search for answers of how to combine AR and VR views and the transitions between them should evaluate when it is more advantageous to use AR or VR and if the complete process could be done in real-time. The VR views could be generated by 3D modelling or 3D reconstruction methods.

In learning, 75% of knowledge comes visually and 13% comes through hearing. So, using the hearing sense to extend the egocentric viewpoint is another path to follow. Audio description of a scene, used for blind or visually impaired people, is like an image of that scene taken from the same viewpoint. The challenge is to merge this multimodal information with regular renderings. This part of the work must start to replicate the visual cues of the off-screen visualization techniques, using audio, when current user's actions implicate reactions in places outside the user's FOV. Another valid

contribution will be the use of audio descriptions of the physical referents when occlusions occur, according to the user's viewpoint.

All this work must be seconded with an analysis of how to integrate and organise the needed augmented information that will be presented in the camera's images. One example of this can be seen in [10], where preliminary steps towards the characterization of visualization design space for field analysis were provided. This situated information must be thought beyond the visible graphics and annotations, considering other kinds of information such as user's profile, other types of context data, or even multimodal information. Solutions for the egocentric viewpoint limitation might also require the adaptation of the scene's information presented in the user's display. The scene adequacy and modification can transform the same application in completely different ways for both situations or users, changing the displayed information towards the user's profile/condition or the degree of sensitivity of the information (displaying or not sensitive data).

All the proposed approaches will be evaluated, using appropriate methods, in different real-world settings, considering several parameters with known methods and new approaches. Ecological validation from the comparison results will be done. In the end, the creation of guidelines regarding the egocentric viewpoint limitation challenge is an expected contribution in some relevant scenarios.

4 CONCLUSION

When specifically thinking in SV based on AR, the solution for the egocentric viewpoint challenge becomes much more imperative. Solutions for this challenge end up not only finding new and different viewpoints but also presenting important data related to the context where the user is. So, the relevance, amount, and source of coherent data that can be used may be crucial. Considering the presented content, solving, or mitigating this challenge will impact many tasks. In future, this path could use better AI, big data techniques, and wearable displays.

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